

Indian Institute of Technology Kharagpur

Department of Physics

Simulation and Analysis of Robo-Matter: Active Magnetic Magbots with Chiral Dynamics and Light-Field Control

Bachelor Thesis Project Report

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Kharagpur, India

Declaration

I certify that the work contained in this report has been carried out by me under the guidance of my supervisor. The work has not been submitted to any other institute for any degree or diploma. I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

Whenever I have used material from other sources, including research papers, theoretical analysis, figures, computational ideas, and text, I have given due credit through citations and references. The simulations, interpretations, and analysis presented in this report are based on the computational work and material collected for this Bachelor Thesis Project.

Date: May 1, 2026

Place: Kharagpur

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Certificate

This is to certify that the project report entitled *Simulation and Analysis of Robo-Matter: Active Magnetic Magbots with Chiral Dynamics and Light-Field Control* submitted by Suraj (Roll No.: 22PH23011) to Indian Institute of Technology Kharagpur towards partial fulfilment of the requirements of the Bachelor Thesis Project is a record of bona fide work carried out under my supervision and guidance during Spring Semester, 2025–26.

The work presented in this report involves the modelling, simulation, and analysis of Robo-Matter inspired by active magnetic Magbot systems. The student has studied the relevant literature, developed physics-based computational models, analyzed collective phase-like behavior, and documented the results in the present thesis report.

Date: May 1, 2026

Place: Kharagpur

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Abstract

Reconfigurable multifunctional materials are an important frontier in modern physics, materials science, and robotics because they seek to combine structural adaptability with active, programmable behavior. Recent experimental work on Robo-Matter has shown that simple robotic building blocks, called Magbots, can exhibit both matter-like properties such as self-assembly, active crystalline order, and phase-like transitions, and robot-like properties such as smart healing, morphing, force output, and infiltration. These behaviors arise because each building block possesses active motion, tunable magnetic interaction, and coupling to an external programmable light field.

This project develops a two-dimensional computational model of Robo-Matter using active magnetic disk particles inspired by Magbots. Each simulated Magbot is represented as a rigid circular body containing six magnetic sites around its perimeter. The particles possess position, body orientation, and internal magnet phase variables. Their dynamics combine self-propulsion, chiral rotation, light-field modulation, Gaussian light-gradient drift, soft steric repulsion, magnetic attraction, body torque, and internal magnet phase torque. A cell-list neighbor search is used to reduce unnecessary pairwise checks and make simulations of 100–150 Magbots computationally tractable.

The central physical question studied in this report is how collective organization emerges from the competition between activation, which agitates the system, and magnetic binding, which promotes local order. By varying activity, magnetic strength, interaction range, density, and noise, the simulation reproduces four representative phase-like regimes: active glass-like, active crystal, liquid-like, and gas-like states. The results are analyzed using local topological order and nearest-neighbor change rate, which together separate static ordering from dynamic rearrangement. The phase plots show that strong magnetic binding with moderate activity supports ordered active-crystal-like behavior, strong binding with insufficient rearrangement produces arrested glass-like behavior, intermediate activity produces liquid-like restructuring, and weak binding with high activity produces gas-like dispersion.

Overall, the work shows that a simplified physics-inspired model can capture several essential features of Robo-Matter: programmable activation, collective ordering, dynamic bond rearrangement, self-organization, and phase-like transitions. The model is not an

exact replica of the experimental system; rather, it is a compact computational platform for understanding the mechanisms by which simple active magnetic agents generate rich collective behavior. The project contributes a clear modelling framework, phase interpretation, parameter analysis, and simulation basis for future studies of adaptive robotic materials.

Contents

Declaration	i
Certificate	ii
Acknowledgement	iii
Abstract	iv
1 Introduction	1
1.1 Background of the Study	1
1.2 Motivation	2
1.3 Problem Statement	2
1.4 Scope of the Study	2
1.5 Report Organization	3
2 Literature Review	4
2.1 Smart and Active Materials	4
2.2 Robo-Matter Concept	4
2.2.1 Matter-Like Behaviors	4
2.2.2 Robot-Like Behaviors	5
2.3 Active Brownian and Chiral Particle Models	5
2.4 Computational Modelling of Robotic Swarms	5
2.5 Research Gap	6
3 Rationale of the Study	7
3.1 Why This Study Matters	7

3.2	Computational Advantages	7
3.3	Core Physical Competition	8
4	Objectives	9
5	Materials and Methods	10
5.1	System Design Overview	10
5.2	State Variables	10
5.3	Light Field	11
5.4	Light-Gradient Drift	11
5.5	Magnetic Interaction	12
5.6	Steric Repulsion and Collision Handling	12
5.7	Equations of Motion	13
5.8	Boundary Conditions	13
5.9	Cell-List Neighbor Search	13
5.10	Analysis Metrics	14
5.10.1	Local Topological Order	14
5.10.2	Nearest-Neighbor Change Rate	14
5.11	Parameter Sets	14
6	Theory	15
6.1	Active Brownian Motion	15
6.2	Chiral Active Motion	15
6.3	Mean Squared Displacement	16
6.4	Activation Versus Binding	16
6.5	Role of Magnet Symmetry	17
6.6	Relation to Experimental Robo-Matter	17
7	Results and Discussion	18
7.1	Overview of Observed Regimes	18
7.2	Active Crystal Phase	18
7.3	Active Glass-Like Phase	19

7.4	Liquid-Like Phase	20
7.5	Gas-Like Phase	21
7.6	Comparison of Four Regimes	22
7.7	Interpretation Using Order and Dynamics	23
7.8	Physical Mechanism of Transitions	24
7.9	Comparison with Experimental Robo-Matter	24
7.10	Strengths of the Model	25
7.11	Limitations of the Model	25
7.12	Discussion	25
8	Conclusion	26
8.1	Summary	26
8.2	Key Findings	26
8.3	Project Contribution	27
8.4	Future Work	27
8.5	Final Remarks	27
	References	28
A	Simulation Parameters	29
A.1	Default Magbot Geometry	29
A.2	Common Numerical Parameters	29
B	Mathematical Derivations	30
B.1	Euler-Maruyama Discretization	30
B.2	Torque from Off-Center Force	30
C	Algorithm and Code Structure	31
C.1	Main Algorithm	31
C.2	Representative Update Pseudocode	31
C.3	Source Files Used	32
D	Phase Notes	33

D.1 Parameter Influence	33
D.2 Transition Summary	33

List of Figures

5.1	Schematic of the simulated Magbot model. Each disk has a body orientation, six magnetic sites, active propulsion, magnetic interactions, and response to a moving light field.	10
7.1	Active crystal regime. The local topological order rapidly becomes high, while the nearest-neighbor change rate decreases after initial rearrangement. This indicates formation of a stable ordered active structure.	19
7.2	Active glass-like regime. The system develops local organization, but the nearest-neighbor change rate decays strongly, indicating arrested dynamics and reduced ability to reorganize.	20
7.3	Liquid-like regime. Local order develops only gradually, while the nearest-neighbor change rate remains substantial. The system has transient structure but persistent rearrangement.	21
7.4	Gas-like regime. The order parameter remains very low, indicating absence of stable collective organization. Motion and weak binding lead to dispersed, disordered behavior.	22
7.5	Comparison of representative simulated Robo-Matter regimes. The plots use local topological order and nearest-neighbor change rate to distinguish ordered, arrested, fluid, and dispersed behavior.	23

List of Tables

5.1	Representative parameter sets for four simulated Robo-Matter regimes. .	14
6.1	Qualitative transition logic used to interpret simulated phases.	17
7.1	Interpretation of simulated phase-like regimes.	18
7.2	Metric-based classification of collective states.	24
A.1	Core geometric parameters used in the simulation.	29
A.2	Common time integration and noise parameters.	29
C.1	Main local files used as project material.	32

1 Introduction

1.1 Background of the Study

Conventional materials are usually designed to perform a fixed function after fabrication. A crystalline solid may be strong, a polymer may be flexible, and a composite may be lightweight, but the internal building blocks of such materials generally do not make decisions, process information, or actively rearrange after deployment. This limitation has motivated the development of smart and active materials, where the material can respond to external stimuli such as light, temperature, mechanical loading, or chemical signals.

Active matter extends this idea further. It consists of many units that consume energy locally and convert it into motion or mechanical work. Examples include bacterial colonies, active colloids, vibrated granular particles, micro-swimmers, and robotic swarms. Unlike passive materials, active matter can organize itself far from equilibrium, generating clusters, vortices, waves, swirls, and coherent collective motion.

Robo-Matter is a recent concept that connects active matter with swarm robotics. In this framework, the material is composed of robotic building blocks that can move, interact, sense environmental signals, and reconfigure. The experimental Magbot system reported by Wang *et al.* demonstrates that small robotic units with magnetic interactions and light-field control can show both matter-like and robot-like behavior [1]. This dual character is the central inspiration for the present computational project.

Central idea

The project treats Robo-Matter as a many-body physics problem: if simple Magbot-like particles are given active motion, chirality, tunable magnetic binding, and light-field activation, then collective phases and material-like behavior can emerge without prescribing global order.

1.2 Motivation

There is growing interest in materials that can adapt, heal, migrate, and change shape. Potential examples include protective surfaces that repair damage, robotic matter that navigates confined environments, microrobotic carriers for targeted transport, and modular systems that assemble into useful shapes on demand. Direct experimental construction of such systems can be expensive and technically challenging. A computational model is therefore valuable because it allows controlled variation of physical parameters and helps isolate the mechanisms responsible for observed behavior.

The motivation of this project is to understand how rich collective dynamics can arise from a minimal set of physically interpretable rules. Instead of modelling every hardware detail of a real Magbot, this work constructs a simplified active magnetic disk system. The simulation preserves the essential ingredients: circular bodies, multiple magnetic sites, chirality, light-dependent activity, torque, steric exclusion, and environmental control.

1.3 Problem Statement

The broad problem addressed in this report is:

How can a physics-based simulation of active magnetic Magbot-like particles reproduce phase-like Robo-Matter behavior, and which model parameters control the transition between ordered, fluid, and dispersed collective states?

This problem requires a model that is simple enough to simulate and interpret, but detailed enough to include both translational and rotational effects. It also requires analysis metrics that distinguish structural order from dynamic rearrangement.

1.4 Scope of the Study

The scope of the project is limited to two-dimensional simulations of rigid circular Magbot-like particles. The study focuses on the following aspects:

- Construction of a particle-level model containing body motion, chirality, internal magnet phases, light-field coupling, and magnetic interactions.
- Simulation of 100–150 interacting Magbots under different activity and magnetic-binding regimes.
- Identification of active glass-like, active crystal, liquid-like, and gas-like collective states.

- Analysis using local topological order and nearest-neighbor change rate.
- Comparison of the computational model with the qualitative mechanisms reported in experimental Robo-Matter literature.

1.5 Report Organization

Chapter 2 reviews the relevant literature on active matter, Robo-Matter, and Magbot systems. Chapter 3 explains the rationale and research gap. Chapter 4 lists the objectives. Chapter 5 presents the model and numerical methods. Chapter 6 develops the theoretical basis of the dynamics and phase competition. Chapter 7 discusses the results. Chapter 8 concludes the report and suggests future work. Appendices provide parameter tables, derivations, algorithmic details, and source-code structure.

2 Literature Review

2.1 Smart and Active Materials

Smart materials respond to external stimuli by changing measurable properties such as shape, stiffness, optical response, or conductivity. Their importance lies in their ability to move beyond the static role of traditional materials. Active matter, by contrast, is defined by the continuous local consumption of energy by its constituents. This energy consumption drives the system away from thermal equilibrium and produces emergent collective states.

Active matter has been reviewed extensively in the context of colloids, bacteria, cell tissues, and synthetic swimmers [2, 3]. A common theme across these systems is that simple individual-level rules can generate large-scale order. Unlike equilibrium matter, where phases are determined by free-energy minimization, active phases are governed by an interplay among drive, interaction, noise, confinement, and dissipation.

2.2 Robo-Matter Concept

Robo-Matter combines ideas from active matter, smart materials, and robot swarms. The experimental system of Wang *et al.* uses micro-robotic building blocks called Magbots [1]. These units are equipped with active rotation, magnetic binding sites, photosensitive control, and interaction with a programmable light field. The system is not simply a robot swarm, because it can also behave like a material with crystal-like, liquid-like, glass-like, and gas-like states. It is also not simply a passive material, because it can process environmental cues and actively reorganize.

2.2.1 Matter-Like Behaviors

The matter-like aspect of Robo-Matter appears in collective self-organization. Magbots can assemble into ordered structures, form active crystals, reorganize under changed activation, and display phase-like transitions. The experimental paper reports active glass-like, active

crystal, liquid-like, and gas-like states, controlled primarily by the competition between activation strength and magnetic binding strength [1].

2.2.2 Robot-Like Behaviors

The robot-like aspect appears when Magbot assemblies respond to programmed light patterns. Reported demonstrations include active force output, smart healing, morphing, infiltration through barriers, cargo search, and robust operation under partial unit failure. These functions require information exchange between the building blocks and the environment. In the experiment, this is achieved through camera tracking, an LED platform, and light-sensitive Magbot response.

2.3 Active Brownian and Chiral Particle Models

The Active Brownian Particle (ABP) model is a standard theoretical description of self-propelled particles. In its simplest form, each particle moves with a constant speed along an orientation vector that undergoes rotational diffusion. Chiral active particles include an additional intrinsic angular velocity, causing curved or circular motion [4]. Such models are useful because they capture persistent motion, orientational decorrelation, ballistic-to-diffusive crossover, and collective effects under interaction.

The present project uses an ABP-inspired framework but extends it in three ways. First, the particles possess multiple magnetic sites instead of simple isotropic attraction. Second, each magnetic site has an internal phase variable that can rotate. Third, the particle activity is coupled to a moving Gaussian light field, mimicking programmable external activation.

2.4 Computational Modelling of Robotic Swarms

Robotic-swarm simulations often emphasize communication, control laws, and task execution. Active-matter simulations emphasize physical interaction, noise, and emergent collective dynamics. Robo-Matter sits between these traditions. For this reason, the present simulation adopts a physics-first approach: it does not prescribe a desired global structure, but allows local forces and torques to produce organization.

Why simulation matters

Experiments demonstrate what Robo-Matter can do. Simulation helps answer why it happens: which terms in the equations promote binding, which terms melt order, how light-field modulation acts like activation, and how torque couples local magnet alignment to whole-body rotation.

2.5 Research Gap

The experimental literature demonstrates Robo-Matter using physical Magbots and a sophisticated feedback platform. However, a beginner-friendly computational model that connects the experimental idea to explicit equations of motion is useful for learning and further exploration. The gap addressed here is the development and interpretation of a compact, physics-based simulation that reproduces key qualitative phase regimes while remaining transparent enough to modify.

3 Rationale of the Study

3.1 Why This Study Matters

Robo-Matter provides a route to materials that can change their structure and function after deployment. Understanding such systems requires a blend of mechanics, statistical physics, nonlinear dynamics, and robotics. A computational study is especially appropriate because the phase behavior depends on many coupled parameters: activity, attraction, density, noise, chirality, confinement, and light control.

This project matters because it transforms the Robo-Matter idea from a qualitative concept into an explicit simulation model. The model allows one to ask: what happens if magnetic attraction is increased? What if the light field is stronger? What if the box is made larger? What if rotation becomes faster? These questions are difficult to answer from static figures alone but can be explored through parameter sweeps.

3.2 Computational Advantages

The simulation approach provides several advantages:

- **Controlled parameters:** each physical mechanism can be varied independently.
- **Repeatability:** identical random seeds allow direct comparison between cases.
- **Observability:** all particle positions, orientations, forces, and magnet phases are available.
- **Efficiency:** cell-list neighbor search avoids unnecessary all-to-all interaction checks.
- **Extendibility:** additional light fields, chirality distributions, boundary conditions, and metrics can be added.

3.3 Core Physical Competition

The central competition in the model is between activation and magnetic binding:

$$\text{collective state} \sim \frac{\text{magnetic binding and ordering}}{\text{active agitation and noise}}. \quad (3.1)$$

When binding dominates, particles stick and order. When activation dominates, bonds break and the system becomes fluid or gas-like. Between these extremes, the system can produce nontrivial dynamic structures.

Interpretive principle

The same interaction rules can generate different material states. The observed phase is not hard-coded; it emerges from the relative strength of activity, attraction, density, and noise.

4 Objectives

The objectives of the project are:

1. To study the concept of Robo-Matter and identify the essential physical mechanisms behind Magbot-based collective behavior.
2. To construct a two-dimensional simulation of active magnetic Magbot-like particles.
3. To include chiral rotation, self-propulsion, light-controlled speed, light-gradient drift, steric repulsion, magnetic attraction, body torque, and internal magnet phase torque.
4. To implement efficient neighbor detection using a cell-list method.
5. To generate and compare representative active glass-like, active crystal, liquid-like, and gas-like regimes.
6. To analyze phase-like behavior using local topological order and nearest-neighbor change rate.
7. To connect the computational observations with experimental Robo-Matter behavior reported in the literature.

5 Materials and Methods

5.1 System Design Overview

The simulation represents Robo-Matter as a collection of rigid circular particles in a square two-dimensional domain. Each particle is a Magbot-like unit with a body center, body angle, and six magnetic sites arranged around its perimeter. The system is evolved using an overdamped discrete-time update, appropriate for a strongly dissipative robotic or granular environment where inertia is not the dominant modelling feature.

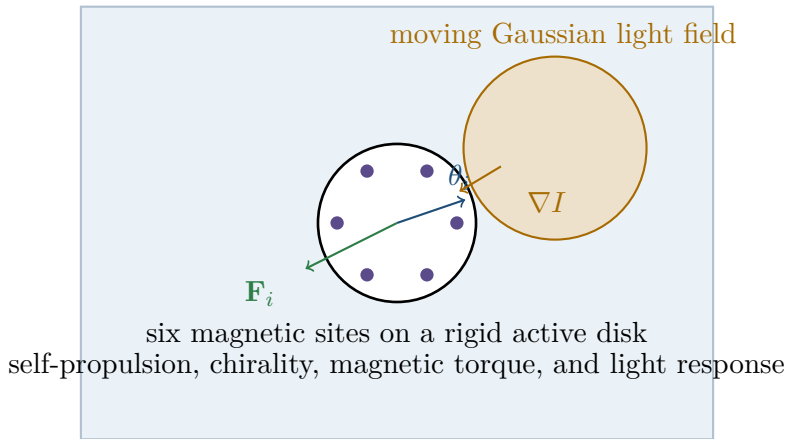


Figure 5.1: Schematic of the simulated Magbot model. Each disk has a body orientation, six magnetic sites, active propulsion, magnetic interactions, and response to a moving light field.

5.2 State Variables

For the i th Magbot, the main state variables are:

$$\mathbf{x}_i(t) = (x_i(t), y_i(t)), \quad \theta_i(t), \quad \phi_{ik}(t), \quad k = 1, \dots, N_{\text{mag}}. \quad (5.1)$$

Here $\mathbf{x}_i$ is the body-center position, θ_i is the body orientation, and ϕ_{ik} is the internal phase of the k th magnetic site. The fixed angular location of the k th site on the body is denoted by α_k . The absolute magnet angle is therefore

$$\psi_{ik} = \theta_i + \alpha_k + \phi_{ik}. \quad (5.2)$$

The six fixed anchor points are placed at radius R_{ring} from the center:

$$\mathbf{a}_k = R_{\text{ring}} \begin{bmatrix} \cos \alpha_k \\ \sin \alpha_k \end{bmatrix}, \quad \alpha_k = \frac{2\pi(k-1)}{N_{\text{mag}}}. \quad (5.3)$$

5.3 Light Field

The simulation uses a Gaussian light field to represent a programmable activation pattern. For a light center $\mathbf{x}_L(t)$, the intensity at position $\mathbf{x}$ is

$$I(\mathbf{x}, t) = I_0 + A_L \exp \left[-\frac{|\mathbf{x} - \mathbf{x}_L(t)|^2}{2\sigma_L^2} \right]. \quad (5.4)$$

For the moving-spot case,

$$\mathbf{x}_L(t) = R_L \begin{bmatrix} \cos(2\pi t/T_L) \\ \sin(2\pi t/T_L) \end{bmatrix}. \quad (5.5)$$

The intensity is clipped to the interval $[0, 1]$ in the implementation. The light affects both translational and rotational motion. It increases the active speed and changes the chiral angular velocity:

$$v_i = v_{\text{base}} + \gamma I_i, \quad (5.6)$$

$$\omega_i = \chi_i (\omega_{\text{base}} + \beta I_i), \quad (5.7)$$

where $\chi_i = -1$ denotes clockwise rotation and $\chi_i = +1$ denotes counterclockwise rotation.

5.4 Light-Gradient Drift

The model includes a simplified drift toward brighter regions:

$$\mathbf{F}_{L,i} = k_L \nabla I(\mathbf{x}_i, t). \quad (5.8)$$

For the Gaussian component,

$$\nabla I = I_{\text{spot}} \frac{\mathbf{x}_L - \mathbf{x}}{\sigma_L^2}. \quad (5.9)$$

This rule is not intended as a full electromechanical model of the real photoresistor-motor coupling. It is a compact way to represent light-guided activation and migration.

5.5 Magnetic Interaction

Each Magbot has six magnetic sites. For a pair of magnetic sites separated by vector $\mathbf{r}$ and distance r , the simulation uses the simplified interaction energy

$$U_{ab} = -k_m \exp\left(-\frac{r^2}{r_0^2}\right) \cos(\Delta\psi), \quad (5.10)$$

where k_m is magnetic strength, r_0 is the decay scale, and $\Delta\psi$ is the relative magnet-angle difference. The interaction is evaluated only when $r < r_{\text{cut}}$.

The force contribution is obtained from the gradient of this potential. In the implemented form,

$$\mathbf{F}_{ab} = \frac{2k_m}{r_0^2} \exp\left(-\frac{r^2}{r_0^2}\right) \cos(\Delta\psi) \mathbf{r}. \quad (5.11)$$

The internal magnet phase torque is

$$\tau_\phi = k_m \exp\left(-\frac{r^2}{r_0^2}\right) \sin(\Delta\psi). \quad (5.12)$$

When a magnetic force acts away from the body center, it also creates body torque:

$$\tau_{\text{body}} = \mathbf{r}_{\text{lever}} \times \mathbf{F}. \quad (5.13)$$

5.6 Steric Repulsion and Collision Handling

To prevent unphysical overlap, a hard-core repulsive force is applied when two disks overlap. If d_{ij} is the center-to-center distance,

$$\mathbf{F}_{\text{rep}} = -k_{\text{rep}}(2R_{\text{bot}} - d_{ij})\hat{\mathbf{r}}_{ij}, \quad d_{ij} < 2R_{\text{bot}}. \quad (5.14)$$

After force-based integration, the code also performs direct overlap resolution through a few sweeps. This improves numerical stability in dense configurations.

5.7 Equations of Motion

The overdamped translational update is

$$\mathbf{x}_i(t + \Delta t) = \mathbf{x}_i(t) + [\mu_t \mathbf{F}_i + \mathbf{F}_{L,i} + v_i \mathbf{n}_i] \Delta t + \sqrt{2D_t \Delta t} \boldsymbol{\eta}_i, \quad (5.15)$$

where $\mathbf{n}_i = (\cos \theta_i, \sin \theta_i)$ and $\boldsymbol{\eta}_i$ is a Gaussian random vector.

The body angle update is

$$\theta_i(t + \Delta t) = \theta_i(t) + [\omega_i + \mu_r \tau_i] \Delta t + \sqrt{2D_r \Delta t} \xi_i. \quad (5.16)$$

The internal magnet phase update is

$$\phi_{ik}(t + \Delta t) = \phi_{ik}(t) + \mu_\phi \tau_{\phi,ik} \Delta t + \sqrt{2D_\phi \Delta t} \zeta_{ik}. \quad (5.17)$$

5.8 Boundary Conditions

The main simulations use reflecting square boundaries. If a particle crosses a vertical wall, its orientation is reflected as

$$\theta \rightarrow \pi - \theta. \quad (5.18)$$

If it crosses a horizontal wall,

$$\theta \rightarrow -\theta. \quad (5.19)$$

The position is clipped inside the simulation box. This mimics elastic reflection at rigid boundaries.

5.9 Cell-List Neighbor Search

Direct pairwise interaction checking scales as $O(N^2)$ and becomes inefficient for large N . The simulation therefore divides the domain into cells whose side length is comparable to the interaction cutoff. Each particle checks only particles in its own cell and neighboring cells. This keeps the interaction search efficient while preserving local pair detection.

Computational design choice

The cell-list method is important because the model contains both body-level and site-level interactions. Without local neighbor filtering, checking all possible site pairs for all Magbot pairs would become unnecessarily expensive.

5.10 Analysis Metrics

The results are analyzed using two complementary metrics.

5.10.1 Local Topological Order

The local topological order parameter measures the degree of local structural organization. A high value indicates a stable neighborhood arrangement or locally ordered structure. In the plots generated for this project, a threshold line near 0.30 separates low-order gas-like or fluid-like states from more ordered regimes.

5.10.2 Nearest-Neighbor Change Rate

The nearest-neighbor change rate n_c measures how frequently the local neighborhood of particles changes. A low value indicates arrested or solid-like behavior, while a high value indicates frequent bond breaking and rearrangement. In the plots, a threshold line near $n_c = 5$ is used as a practical separator between relatively stable and actively reorganizing structures.

5.11 Parameter Sets

Representative parameter choices used in the project are summarized in Table 5.1. These values are extracted from the simulation scripts in the `states` folder.

Table 5.1: Representative parameter sets for four simulated Robo-Matter regimes.

Regime	N	box	v_{base}	ω_{base}	k_m	r_0
Active glass-like	150	20.0	0.45	0.50	5.80	4.00
Active crystal	150	17.0	10.45	6.90	10.00	5.00
Liquid-like	100	20.0	6.50	5.80	2.50	0.95
Gas-like	100	28.0	10.00	10.00	0.80	0.80

6 Theory

6.1 Active Brownian Motion

A minimal active Brownian particle has the dynamics

$$\dot{\mathbf{x}} = v_0 \mathbf{n}(t) + \sqrt{2D_t} \boldsymbol{\eta}(t), \quad (6.1)$$

$$\dot{\theta} = \sqrt{2D_r} \xi(t). \quad (6.2)$$

At short times, the particle keeps its direction and motion is approximately ballistic. At long times, rotational diffusion destroys directional memory and the motion becomes diffusive. This crossover is central to active matter.

6.2 Chiral Active Motion

For a chiral particle, an intrinsic angular velocity is added:

$$\dot{\theta} = \omega + \sqrt{2D_r} \xi(t). \quad (6.3)$$

If noise and interaction are absent, the particle follows a circular trajectory with approximate radius

$$R_{\text{circle}} = \frac{v_0}{|\omega|}. \quad (6.4)$$

In the Robo-Matter simulation, chirality is important because Magbots in the experimental system rotate due to vibration motors and tilted brushes. The sign of ω controls clockwise or counterclockwise motion.

6.3 Mean Squared Displacement

Although the primary project plots focus on structural order and neighbor-change rate, the theoretical mean squared displacement (MSD) remains a useful reference quantity. For a non-chiral ABP in two dimensions,

$$\langle \Delta r^2(t) \rangle = 4D_t t + \frac{2v_0^2}{D_r^2} (D_r t + e^{-D_r t} - 1). \quad (6.5)$$

For $t \ll D_r^{-1}$, the leading active contribution is approximately $v_0^2 t^2$, indicating ballistic motion. For $t \gg D_r^{-1}$, the MSD becomes diffusive with an effective diffusion coefficient

$$D_{\text{eff}} = D_t + \frac{v_0^2}{2D_r}. \quad (6.6)$$

In interacting Robo-Matter, this simple expression is modified by magnetic binding, confinement, collision, and collective rearrangement. However, the ballistic-to-diffusive idea remains useful for interpreting activity-driven spreading.

6.4 Activation Versus Binding

The phase behavior can be interpreted through a dimensionless qualitative ratio:

$$\Lambda = \frac{k_m}{v_{\text{base}} + \omega_{\text{base}} + \epsilon}, \quad (6.7)$$

where ϵ prevents division by zero and reminds us that this is not a rigorous thermodynamic parameter. Large Λ indicates strong binding compared with agitation. Small Λ indicates weak binding compared with activity. In the project, this ratio is used conceptually rather than as a fitted universal order parameter.

Table 6.1: Qualitative transition logic used to interpret simulated phases.

Condition	Expected behavior	Representative phase
Strong binding, low activity	Particles bind but cannot easily escape disordered local traps.	Active glass-like
Strong binding, moderate activity	Motion anneals defects while attraction maintains local order.	Active crystal
Comparable binding and activity	Bonds form and break continually; the network flows.	Liquid-like
Weak binding, high activity	Particles or small clusters roam independently.	Gas-like

6.5 Role of Magnet Symmetry

Six magnetic sites produce a natural route to local hexagonal or honeycomb-like packing. The fixed body anchors introduce anisotropy: attraction depends not only on distance but also on relative magnet orientation. This makes the model richer than a simple isotropic Lennard-Jones-like attraction. Body torque and internal magnet torque allow local magnetic alignment to influence both individual particle orientation and collective structure.

6.6 Relation to Experimental Robo-Matter

The experimental paper reports that homogeneous light intensity can play a temperature-like role: stronger light increases Magbot activation and changes collective organization [1]. In this simulation, the light field increases active speed and rotation rate. Therefore, activity can melt stable clusters, promote reorganization, or disperse the system depending on magnetic binding strength.

7 Results and Discussion

7.1 Overview of Observed Regimes

The simulation produces four representative phase-like regimes. They are not thermodynamic phases in the strict equilibrium sense; they are nonequilibrium collective states distinguished by local order, bond rearrangement, density, and activity.

Table 7.1: Interpretation of simulated phase-like regimes.

Regime	Structural signature	Dynamical signature
Active glass-like	Locally connected or jammed structure with limited ability to rearrange globally.	Neighbor-change rate decays; motion becomes arrested.
Active crystal	High local order and stable neighborhood structure.	Defects are annealed; neighbor changes fall below threshold.
Liquid-like	Partial local order with continuous restructuring.	Neighbor changes remain frequent; bonds form and break.
Gas-like	Low topological order and mostly separated particles or small clusters.	Particles remain mobile and disordered.

7.2 Active Crystal Phase

The active crystal simulation uses strong magnetic binding and sufficiently high motion to allow reorganization. Figure 7.1 shows that the local topological order rises rapidly from a low value to above 0.8 and remains high for most of the simulated time. The nearest-neighbor change rate initially spikes, corresponding to active rearrangement and defect correction, but then decreases below the threshold line.

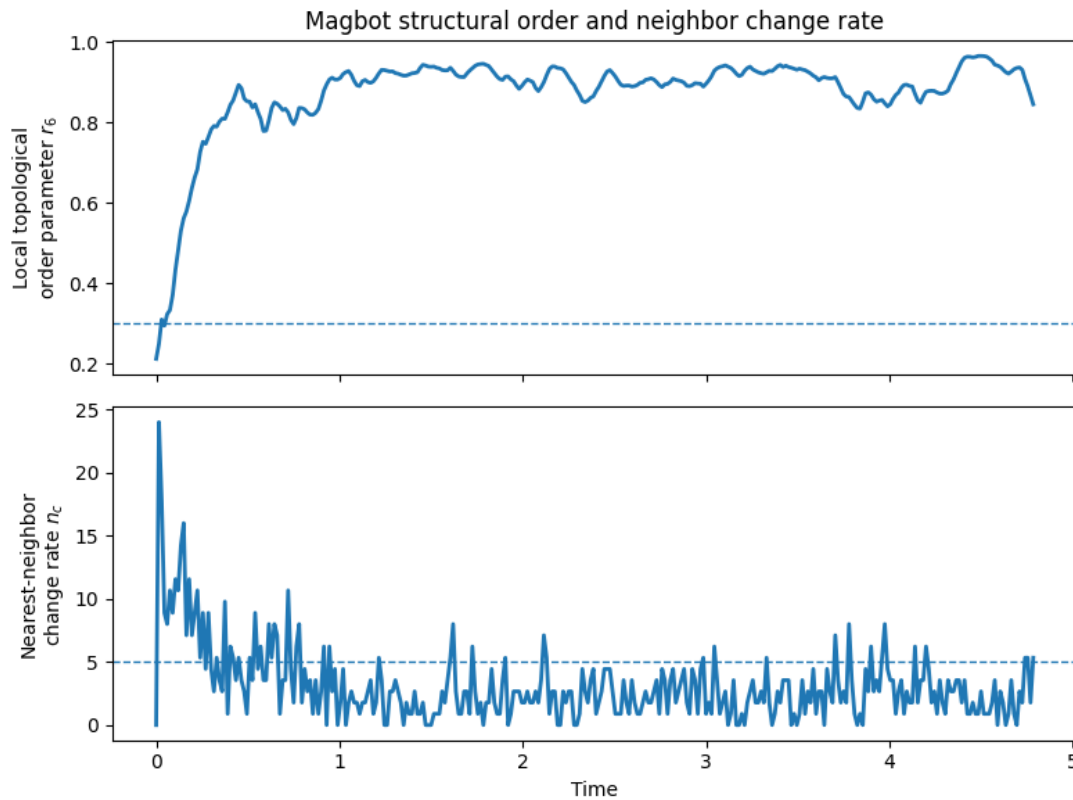


Figure 7.1: Active crystal regime. The local topological order rapidly becomes high, while the nearest-neighbor change rate decreases after initial rearrangement. This indicates formation of a stable ordered active structure.

Result highlight: active crystal

The active crystal is produced when magnetic binding is strong but the system still has enough motion to anneal defects. In physical terms, activity is not merely destructive; at moderate levels it helps the system escape local disorder and settle into a more ordered state.

7.3 Active Glass-Like Phase

The active glass-like case corresponds to strong attraction and low activity. Figure 7.2 shows a gradual increase in order and a decay of neighbor changes. The key distinction from an ideal active crystal is that rearrangement becomes suppressed before the system can fully explore configuration space. This is characteristic of a kinetically arrested state.

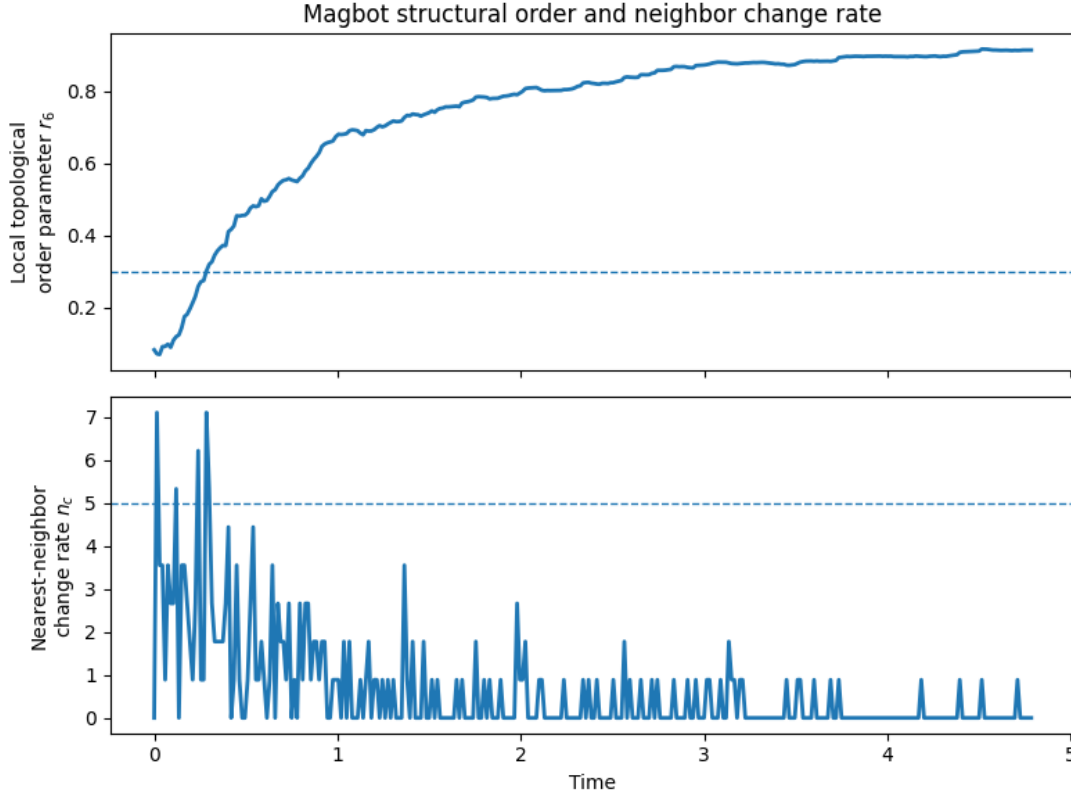


Figure 7.2: Active glass-like regime. The system develops local organization, but the nearest-neighbor change rate decays strongly, indicating arrested dynamics and reduced ability to reorganize.

In the Robo-Matter interpretation, the glass-like state is not simply “low order.” It may contain local order but lacks sufficient rearrangement to remove defects or produce a clean crystalline structure. This agrees with the general physical idea that glassiness is associated with kinetic arrest.

7.4 Liquid-Like Phase

The liquid-like simulation uses activity comparable to magnetic binding. Figure 7.3 shows that local order grows gradually and eventually crosses the practical threshold, but the neighbor-change rate remains large and fluctuating. This is the defining signature of liquid-like behavior: local coordination exists, but bonds do not remain fixed.

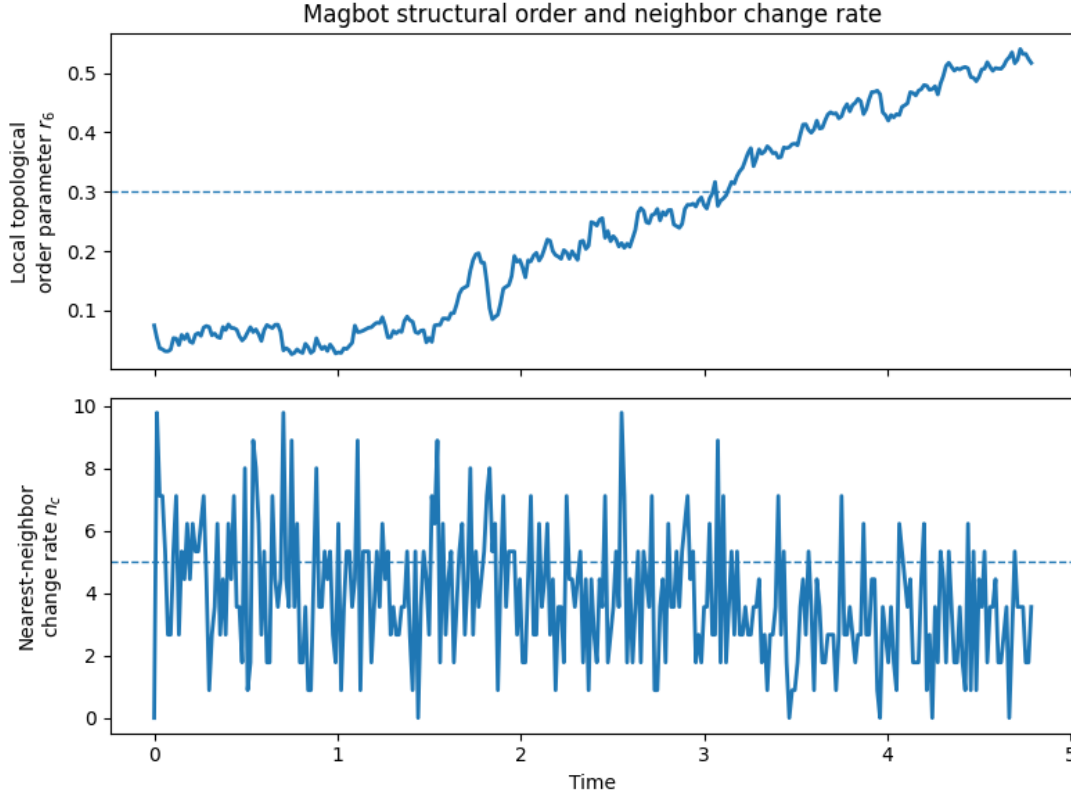


Figure 7.3: Liquid-like regime. Local order develops only gradually, while the nearest-neighbor change rate remains substantial. The system has transient structure but persistent rearrangement.

The liquid-like state is important because it demonstrates that collective organization need not be static. The particles can form temporary clusters and networks while continuously exchanging neighbors. This is similar to ordinary liquids, but here the motion is sustained by active drive rather than thermal equilibrium alone.

7.5 Gas-Like Phase

In the gas-like regime, activity is high and magnetic attraction is weak. Figure 7.4 shows that the local topological order remains near zero and well below the threshold. Neighbor-change rate fluctuates because particles continue to encounter and separate from one another, but sustained binding is absent.

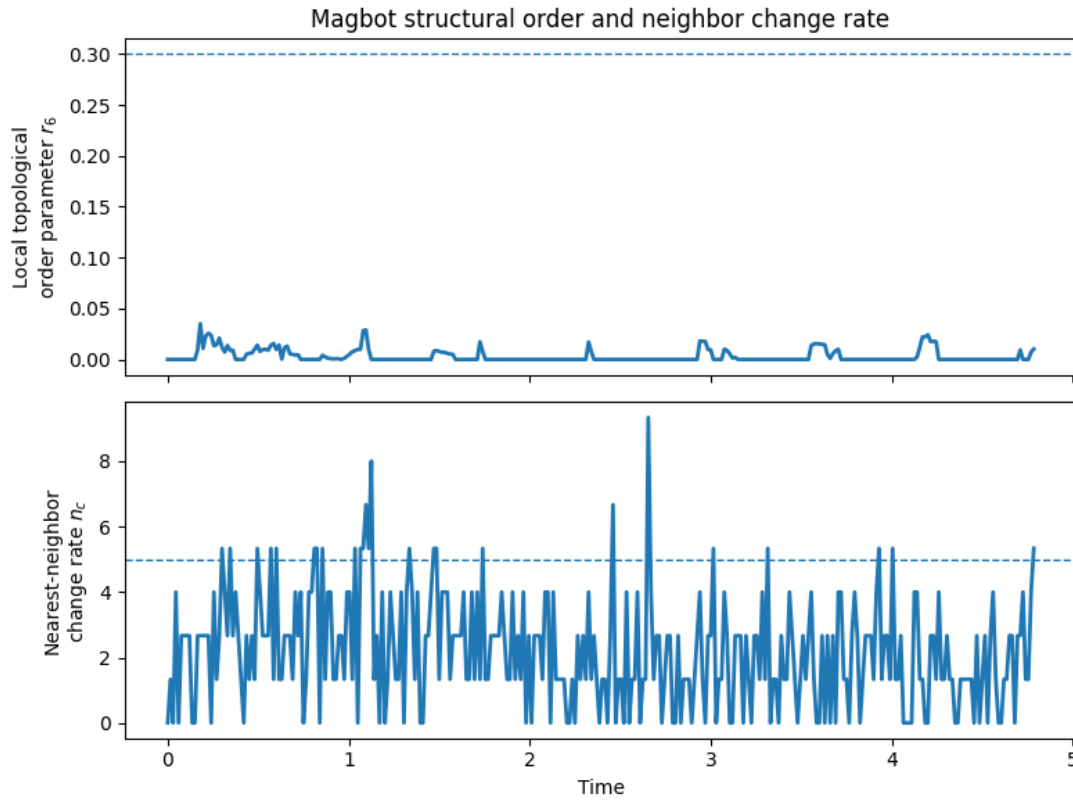


Figure 7.4: Gas-like regime. The order parameter remains very low, indicating absence of stable collective organization. Motion and weak binding lead to dispersed, disordered behavior.

7.6 Comparison of Four Regimes

Figure 7.5 places the four generated plots in a single visual comparison. This panel summarizes the main simulation result: changing the balance between activity, binding, and density changes the emergent Robo-Matter state.

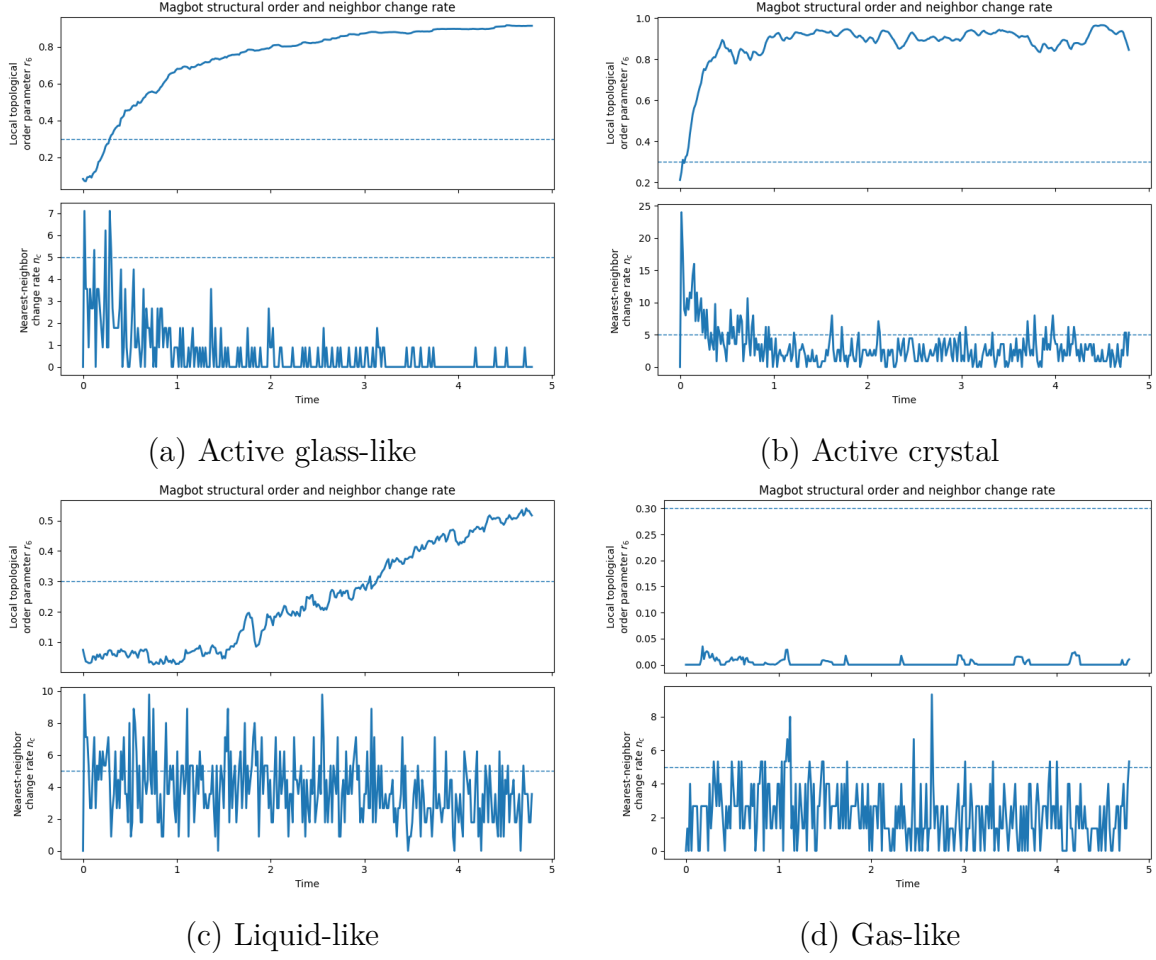


Figure 7.5: Comparison of representative simulated Robo-Matter regimes. The plots use local topological order and nearest-neighbor change rate to distinguish ordered, arrested, fluid, and dispersed behavior.

7.7 Interpretation Using Order and Dynamics

The two selected metrics should be interpreted together. A high order parameter alone does not prove that the system is a good crystal, because a jammed state may also have local order. Similarly, a high neighbor-change rate alone does not prove disorder, because a liquid may have local structure while still rearranging. The useful classification is obtained by combining them:

Table 7.2: Metric-based classification of collective states.

Metric combination	Physical meaning	Likely regime
High order, low n_c	Stable local structure with little rearrangement.	Active crystal or arrested glass
High/medium order, high n_c	Structured but flowing network.	Liquid-like
Low order, high/medium n_c	Disordered mobile particles or small clusters.	Gas-like
Low order, low n_c	Frozen disordered state or insufficient motion.	Deep glass-like limit

7.8 Physical Mechanism of Transitions

The transition sequence can be understood as:

$$\begin{array}{c}
 \text{glass-like} \xrightarrow{\text{increase activity}} \text{active crystal} \xrightarrow{\text{increase activity further}} \text{liquid-like} \\
 \text{weak binding or high activity} \xrightarrow{\hspace{1.5cm}} \text{gas-like.}
 \end{array} \tag{7.1}$$

The first transition is subtle. A small increase in activity can improve order by allowing particles to escape trapped local configurations. However, too much activity breaks bonds faster than they can stabilize, producing liquid-like or gas-like behavior.

7.9 Comparison with Experimental Robo-Matter

The computational results are qualitatively consistent with the experimental Robo-Matter phase concept. In the experimental system, homogeneous light intensity affects Magbot rotation speed and plays a temperature-like role, while magnetic binding stabilizes assemblies [1]. The present simulation reproduces the same conceptual competition. It also includes a moving light field, allowing light-guided drift and activity modulation.

Important differences remain. The simulation does not model motor electronics, brush friction, real magnetic rod geometry, camera feedback, LED pixel control, or exact contact mechanics. Therefore, numerical values should not be interpreted as direct experimental predictions. The value of the model lies in mechanism-level understanding.

7.10 Strengths of the Model

The model has several strengths:

- It is physically interpretable: every term in the update rule corresponds to activity, force, torque, noise, or environmental control.
- It contains both translational and rotational dynamics.
- It includes internal magnet phase dynamics, not only fixed attraction.
- It can reproduce four distinct collective regimes using parameter changes.
- It uses efficient local neighbor search.
- It is easy to extend for future parameter sweeps or feedback control.

7.11 Limitations of the Model

The present model also has limitations:

- The magnetic potential is simplified and not a full dipole-field calculation.
- Friction, vibration motor dynamics, and contact mechanics are represented only phenomenologically.
- The light-field drift is a control approximation rather than a detailed photo-electromechanical model.
- The order metrics are useful but not sufficient for complete phase classification.
- Long-time statistical averaging and multiple random seeds are needed for stronger quantitative claims.

7.12 Discussion

The main scientific outcome of the project is the demonstration that Robo-Matter-like behavior can be generated from simple local rules. The simulations support the idea that active robotic materials should be understood as nonequilibrium many-body systems. Activity is not only a source of disorder; it can also enable reconfiguration, defect repair, and transition between states. Magnetic binding is not only a source of stability; if too strong relative to motion, it can arrest the system in a glass-like configuration. The useful operating regime for adaptive material behavior lies between these extremes.

8 Conclusion

8.1 Summary

This project developed a two-dimensional simulation framework for Robo-Matter inspired by active magnetic Magbot systems. Each Magbot was represented as a rigid circular disk with six magnetic sites, chiral rotation, active translational motion, light-field modulation, magnetic force, internal magnet torque, body torque, noise, steric repulsion, and reflecting boundaries. The simulation was implemented in Python and used cell-list neighbor search for efficiency.

The report followed the structure of a formal thesis study: introduction, literature review, rationale, objectives, methods, theory, results, discussion, conclusion, references, and appendices. The core results were organized around four phase-like regimes: active glass-like, active crystal, liquid-like, and gas-like.

8.2 Key Findings

1. Strong magnetic binding and moderate activity support ordered active-crystal-like behavior.
2. Strong binding with insufficient rearrangement produces arrested glass-like behavior.
3. Activity comparable to binding produces liquid-like states with persistent neighbor exchange.
4. Weak binding and high activity produce gas-like dispersed behavior.
5. Local topological order and nearest-neighbor change rate together provide a useful diagnostic pair.
6. The model captures the central Robo-Matter competition between activation and magnetic ordering.

8.3 Project Contribution

The project contributes a transparent computational platform for studying Robo-Matter. It connects experimental ideas from Magbot-based smart materials with explicit equations of motion and interpretable simulation outputs. The work can serve as a foundation for future studies involving feedback control, automated phase diagrams, collective transport, and adaptive morphing.

8.4 Future Work

Future work can improve and extend the project in several directions:

- Perform systematic parameter sweeps over k_m , v_{base} , ω_{base} , density, and noise.
- Compute additional order parameters such as bond-orientational order ψ_6 , cluster-size distribution, radial distribution function, and MSD.
- Use multiple random seeds to obtain statistically reliable phase boundaries.
- Add feedback-controlled light fields that respond to particle positions.
- Model twofold, threefold, and sixfold magnetic-site symmetries separately.
- Study mixed chirality systems containing both clockwise and counterclockwise Magbots.
- Compare simulated phase diagrams more directly with experimental Robo-Matter data.

8.5 Final Remarks

Robo-Matter is a powerful example of how material behavior can emerge from active, interacting building blocks. This project shows that even a simplified simulation can reproduce meaningful collective regimes and explain them through the balance of activity, magnetic binding, and dynamic rearrangement. The work therefore provides both a computational model and a conceptual bridge between active matter physics and reconfigurable robotic materials.

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- [7] Project source material, “Robo-Matter simulation notes and Magbot modelling document,” local file `robo_matter.pdf`, BTP project folder, 2026.
- [8] Project simulation scripts, local files in `states/` and `updated/`, BTP project folder, 2026.

A Simulation Parameters

A.1 Default Magbot Geometry

Table A.1: Core geometric parameters used in the simulation.

Parameter	Value	Meaning
N_{bots}	100–150	Number of simulated Magbots
N_{mag}	6	Magnetic sites per Magbot
R_{bot}	1.0	Radius of each circular body
R_{ring}	0.78	Radius of the magnetic-site ring
r_{mag}	0.105	Visual/display radius of magnetic sites

A.2 Common Numerical Parameters

Table A.2: Common time integration and noise parameters.

Parameter	Typical value	Meaning
D_t	0.012	Translational noise strength
D_r	0.025	Rotational noise strength
D_ϕ	0.012	Internal magnet phase noise
μ_t	1.0	Translational mobility
μ_r	0.50–0.55	Rotational mobility
μ_ϕ	1.0	Magnet phase mobility
Δt	0.003–0.015	Time step
steps	900–1600	Number of frames/updates
seed	7	Random seed used in representative runs

B Mathematical Derivations

B.1 Euler-Maruyama Discretization

For a stochastic differential equation

$$dX = a(X, t) dt + b dW_t, \quad (\text{B.1})$$

the Euler-Maruyama update is

$$X_{n+1} = X_n + a(X_n, t_n)\Delta t + b\sqrt{\Delta t}\eta_n, \quad (\text{B.2})$$

where η_n is a standard normal random variable. This is the basis for the translational, rotational, and internal magnet phase updates used in the simulation.

B.2 Torque from Off-Center Force

For a force $\mathbf{F} = (F_x, F_y)$ applied at lever arm $\mathbf{r} = (r_x, r_y)$ from the center of a two-dimensional rigid body, the out-of-plane torque is

$$\tau_z = r_x F_y - r_y F_x. \quad (\text{B.3})$$

This expression is implemented through the `cross2d` helper function in the project scripts.

C Algorithm and Code Structure

C.1 Main Algorithm

The simulation follows the computational sequence below:

1. Initialize Magbot positions, body orientations, chirality values, and internal magnet phases.
2. At each time step, compute the current light intensity and light gradient.
3. Build a cell list for efficient neighbor detection.
4. For nearby Magbot pairs, compute steric repulsion, magnetic forces, body torques, and internal magnet torques.
5. Update positions, body angles, and internal magnet phases using Euler-Maruyama integration.
6. Apply reflecting boundary conditions and overlap resolution.
7. Update visualization elements including bodies, magnets, heading lines, light field, and interaction lines.
8. Store or plot order metrics for phase analysis.

C.2 Representative Update Pseudocode

Listing C.1: Simplified pseudocode representing the update logic used in the simulation.

```
1 for frame in range(steps):  
2     t = frame * dt  
3  
4     I, gradI = light_intensity_and_gradient(pos, t)  
5     F, T_body, T_phi = compute_forces_and_torques(pos, theta, phi  
        )
```

```

6
7     n = np.column_stack((np.cos(theta), np.sin(theta)))
8     v = v_base + v_light_gain * I
9     omega = chirality * (omega_base + omega_light_gain * I)
10
11     F_light = light_drift_strength * gradI
12
13     pos += (mu_t * F + F_light + v[:, None] * n) * dt
14     pos += np.sqrt(2.0 * Dt * dt) * rng.normal(size=pos.shape)
15
16     theta += (omega + mu_r * T_body) * dt
17     theta += np.sqrt(2.0 * Dr * dt) * rng.normal(size=len(theta))
18
19     phi += mu_phi * T_phi * dt
20     phi += np.sqrt(2.0 * Dphi * dt) * rng.normal(size=phi.shape)
21
22     apply_reflecting_boundaries(pos, theta)
23     resolve_overlaps(pos)

```

C.3 Source Files Used

Table C.1: Main local files used as project material.

File	Role in the report
reference_report.pdf	Structure, front-matter style, and thesis organization.
robo_matter.pdf	Explanation of Magbot variables, forces, torque, light field, and parameter meaning.
Research Papers/s41467-024-53123-6...pdf	Primary research paper on experimental Robo-Matter.
states/active_crystal_phase.py	Parameter set and code basis for active crystal simulation.
states/active_glass_like.py	Parameter set and code basis for glass-like simulation.
states/liquid_like.py	Parameter set and code basis for liquid-like simulation.
states/gas_like_phase.py	Parameter set and code basis for gas-like simulation.
states/*.png	Figures included in the results section.

D Phase Notes

D.1 Parameter Influence

- Increasing k_m promotes binding and local order.
- Increasing r_0 increases the range over which magnetic sites influence each other.
- Increasing v_{base} increases translational agitation.
- Increasing ω_{base} increases chiral rotation and can destabilize bonds if too large.
- Increasing box size reduces density and suppresses frequent clustering.
- Increasing noise tends to destroy order and promote gas-like behavior.

D.2 Transition Summary

Final phase summary

The simulated Robo-Matter states are controlled by the balance between magnetic binding and active agitation. Strong binding with enough rearrangement produces active crystals; strong binding with too little rearrangement produces glass-like arrest; comparable binding and activity produces liquid-like flow; weak binding under high activity produces gas-like dispersion.